

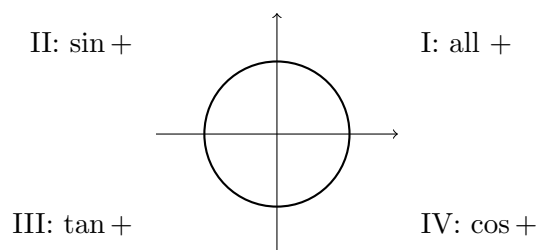
Unit Circle and Trigonometric Graphs — Diagnosing Degree-Radian and Quadrant-Sign Errors

Precalculus

Two errors account for most wrong unit-circle answers, and both produce a clean-looking number:

- **Mode error** — a bare number is read in the wrong angle unit (a calculator left in radian mode, or a conversion run the wrong way).
- **Quadrant-sign error** — the reference angle's value is copied without the sign the quadrant demands.

Give exact values (no decimals) unless a problem asks for a decimal, and state the quadrant you used. Show the reference angle in your work.



Which functions are positive in each quadrant. No values shown — use the angle given in each problem.

1. Warm up. Give the exact value of $\sin \frac{7\pi}{6}$. Name the reference angle and the quadrant before you write the value.

Answer: _____

2. Find and fix. Reva needs $\sin 30^\circ$. She types `sin(30)` into a calculator that is set to radians and reports $\sin 30^\circ \approx -0.99$.

Explain what the calculator actually computed, then give the exact value of $\sin 30^\circ$.

Answer: _____

3. Find and fix. Ari evaluates $\cos \frac{4\pi}{3}$. He writes: “the reference angle is $\frac{\pi}{3}$ and $\cos \frac{\pi}{3} = \frac{1}{2}$, so the answer is $\frac{1}{2}$.”

Name the step Ari skipped, then give the exact value of $\cos \frac{4\pi}{3}$.

Answer: _____

4. Find and fix. Sena evaluates $\tan 150^\circ$ and reports ≈ 0.58 , the value of the tangent of its reference angle.

State the quadrant of 150° and the sign the tangent must carry there, then give the exact value of $\tan 150^\circ$.

Answer: _____

5. Find and fix. Bo converts 225° to radians by multiplying by $\frac{180}{\pi}$ and gets about 12891.6.

Explain how the size of that answer alone shows the conversion ran the wrong way, then give the exact radian measure of 225° .

Answer: _____

6. Solve $2 \sin \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Cal reports $\theta = 30^\circ$ and $\theta = 150^\circ$. He found the reference angle correctly and then placed it in the two quadrants where the sine is *positive*. Give the correct solutions, in degrees.

Answer: _____

7. Solve $2 \cos x - \sqrt{2} = 0$ for $0 \leq x < 2\pi$. Give exact radian answers.

State how many solutions you expect before solving, and which quadrants they lie in.

Answer: _____

8. Challenge. A student writes “ $\cos 3 = 0.9986$ ”.

- (a) Explain what two different angles the bare number 3 could mean, and what the student assumed it meant.
- (b) Compute $\cos 3$ correctly, with 3 measured in radians, rounded to two decimal places.
- (c) Say which quadrant the angle 3 radians lies in, and explain how that quadrant alone tells you the sign of your answer in part (b).

Answer: _____